Maitreyi Tiwari

Indian Institute of Science
Bengaluru, Karnataka

EDUCATION

Indian Institute of Science **2025 - present**
M.Tech - Computational and Data Science ***CGPA - 8.6*** *Bengaluru, Karnataka*

GB Pant University of Agriculture and Technology **2021 - 2025**
B.Tech - Computer Engineering ***CGPA - 8.10*** *Pantnagar, Uttarakhand*

Cynthia Senior Secondary School **2021**
Intermediate - ***Percentage - 97.2%*** *Haldwani, Uttarakhand*

Cynthia Senior Secondary School **2019**
High School - ***Percentage - 99%*** *Haldwani, Uttarakhand*

COURSEWORK / SKILLS

Machine Learning, Edge-AI, NLP, Operating Systems, Computer Networks, Compiler Design, Linear Algebra, Computer Architecture, Python, C, SQL, Git, Database Management System (DBMS), Data Structures & Algorithms, Probability & Statistics

INTERNSHIP

Honeywell Technologies **05-2026 – 08-2026**
AI Intern *Bengaluru*

- Designed and implemented an LLM-assisted Databricks/Spark pipeline to transform highly normalized Building Management System (BMS) schemas across different domains into denormalized Gold views using EAV classification, pivoting, PK/FK extraction, relationship graphs, and automated join planning.
- Developed a data-driven semantic ontology layer capturing column definitions, business rules, glossary terms, enum mappings, and relationships, and integrated 1,300+ semantic facts with MiniLM embeddings and Neo4j for GraphRAG-based, business-aware retrieval.
- Fine-tuned an on-prem Gemma-2-9B-IT Small Language Model using two-stage 4-bit QLoRA on Spark-execution-verified Question-SQL pairs, and built an agentic Text-to-SQL pipeline with semantic retrieval, candidate generation, SQL validation, error-driven repair, execution voting, and value grounding.
- Achieved 80% overall SQL accuracy, 96% execution success, and 95% accuracy on queries across a 50-question benchmark, while serving the 9B model on a single GPU with ∼15 seconds end-to-end latency.

National Informatics Centre Services Inc. **01-07-2024 – 09-08-2024**
Machine Learning Intern *Delhi*

- Developed a real-time object counting system that identifies individuals within predefined zones of interest in a video feed.
- It aimed to manage and monitor population density for enhancing safety, preventing casualties, resource optimization, and enhancing user experiences

PROJECTS

Multi-Label Image Classification & Semantic Segmentation | PyTorch, ResNet, DeepLabV3+, OpenCV

- Developed a 20-class multi-label image classification model using an ImageNet-pretrained ResNet101 backbone with a custom BatchNorm-ReLU-Dropout classification head, trained using AdamW, differential learning rates, focal loss, MixUp, and class-specific threshold optimization.
- Designed and trained a custom DeepLabV3+ variant with a ResNet50 encoder, Atrous Spatial Pyramid Pooling (ASPP), and high-resolution feature fusion for pixel-level semantic segmentation, using a combined Cross-Entropy and Tversky loss to address foreground-background imbalance.
- Applied multi-scale and horizontal-flip test-time augmentation to improve inference robustness, and validated segmentation performance using per-image mIoU, Wilcoxon signed-rank testing, and bootstrap confidence intervals.

- Achieved a test-set mean IoU of 63.97% with a 95% bootstrap confidence interval of [61.57%, 66.40%], with statistically significant improvement over the ResNet50 baseline.

LLM Fine-Tuning & Alignment for Mathematical Reasoning | PyTorch, SFT, Transformers

- Fine-tuned a compact GPT-2-based language model on GSM8K-style mathematical reasoning data using supervised fine-tuning with response-only loss masking and causal language modeling.
- Implemented Direct Preference Optimization (DPO) using chosen/rejected reasoning pairs, a frozen reference model, implicit rewards, and preference-based loss optimization to improve response quality.
- Implemented Group Relative Policy Optimization (GRPO) with group-based reward normalization, relative advantages, importance sampling, clipped policy updates, and KL regularization, and evaluated models using Pass@1 and Pass@K.

Blood Group Detection using Fingerprints | TensorFlow, OpenCV, Machine Learning

- Developed a Convolutional Neural Network (CNN) to predict human blood groups from fingerprint images using supervised learning techniques.
- Used OpenCV for preprocessing fingerprint images, including noise reduction and contrast enhancement to improve model performance.
- Explored a non-invasive approach for blood group identification using biometric fingerprint patterns.

Content-Based Movie Recommendation System | Python, NLP, Scikit-learn

- Developed a content-based movie recommendation system using the TMDB dataset of 5,000+ movies, leveraging CountVectorizer and cosine similarity to generate recommendations based on genres, keywords, cast, crew, and plot.
- Built an end-to-end NLP pipeline involving JSON data handling, text preprocessing with Porter Stemmer, feature engineering, and similarity matrix computation to generate semantically relevant movie recommendations.

Credit Card Fraud Detection | Python, Scikit-learn, Random Forest, SMOTE

- Developed and compared Logistic Regression, Decision Tree, and Random Forest classifiers for credit card fraud detection on a highly imbalanced dataset with a 0.17% fraud rate.
- Implemented data preprocessing and class balancing using StandardScaler, duplicate removal, and SMOTE, and evaluated models using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Binary Sentiment Analyzer for Tweets | Python, NLTK, Scikit-learn, Logistic Regression

- Developed a binary sentiment classifier using Logistic Regression on 1.6 million tweets from the Sentiment140 dataset to classify tweets as positive or negative.
- Implemented text preprocessing using regular expression cleaning, stemming, stopword removal, and TF-IDF vectorization, achieving 78–79% accuracy on both training and testing datasets with balanced classes.

Speech Emotion Recognition | Python, Keras, TensorFlow, Scikit-learn

- Developed a speech emotion classification model to recognize 7 emotions—angry, disgust, fear, happy, neutral, sad, and surprised—using the TESS (Toronto Emotional Speech Set) dataset containing 5,600 audio files.
- Extracted MFCC (Mel-Frequency Cepstral Coefficient) features from 3-second speech clips and trained an LSTM-based neural network with dropout regularization for emotion classification.
- Trained the model for 100 epochs using an 80/20 train-test split and evaluated performance using accuracy, classification reports, and confusion matrices across all 7 emotion classes.

Stock Price Prediction Web Application | Python, TensorFlow, LSTM, Streamlit, yfinance

- Developed a full-stack stock price prediction web application using Streamlit and TensorFlow LSTM networks, enabling users to forecast stock prices up to 52 weeks ahead using real-time market data from yfinance.

- Implemented an end-to-end machine learning pipeline for multiple stocks, including historical data preprocessing, feature engineering with Simple Moving Averages (SMA), LSTM model training, and interactive visualization of actual versus predicted prices.
- Designed a user-friendly predictive analytics interface with a 70/30 train-test split, optimized batch predictions with progress tracking, and integrated data processing, deep learning, and web deployment into a single application.

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Offline Multi-Agent System via Fine-Tuned SLM | Python, LangGraph, MCP, vLLM, PEFT (LoRA), Hugging Face

- Fine-tuned a 1.5B parameter Small Language Model (SLM) using LoRA to serve as a reliable, fully offline tool-caller, achieving 100% schema validity and 93% correct tool selection without cloud API dependencies.
- Architected an on-device multi-agent framework utilizing LangGraph (Supervisor pattern) and the Model Context Protocol (MCP), routing user queries to specialized local agents (Repo, Ops, Docs).
- Implemented dynamic hot-swapping of agent-specific LoRA adapters on a shared base model via vLLM, and engineered an automated synthetic data pipeline to execution-validate tool calls prior to training.

TinyML Head Gesture Recognition System | Arduino, TensorFlow Lite, MicroPython, UDP

- Designed a cloud-free wearable assistive communication device for individuals with severe motor impairments, translating 6 distinct head gestures into real-time OLED text and buzzer alerts.
- Engineered a 12-axis dual-IMU architecture using bilateral Arduino Nicla Vision boards to capture complex asymmetrical movements, streaming data via a custom 50 Hz Wi-Fi UDP pipeline
- Built a feature engineering pipeline extracting 148 statistical, spectral, and cross-IMU symmetry features across 1-second sliding windows to reliably distinguish directional head tilts.
- Trained and deployed an INT8-quantized Keras MLP and an optimized m2cgen Decision Tree directly onto an STM32 microcontroller, achieving 100% test accuracy and 15–40 ms on-device inference latency.

Curvature-Guided Rewiring for Over-Squashing in Hypergraph Neural Networks |
Python, PyTorch, NumPy, SciPy | *M.Tech Thesis, Ongoing*

- Investigating over-squashing in Hypergraph Neural Networks (HGNNs) and proposing curvature-guided hypergraph rewiring as a purely structural remedy that requires no architectural changes.
- Empirically demonstrated over-squashing in VC-HNNs (DeepHGNN) and VH-HNNs (AllSetTransformer) via controlled depth sweeps on CoauthorshipCora, showing accuracy degradation of up to 8.6 points at depth L=4, and confirmed via synthetic YadatiPath and UniformPath benchmarks with ground-truth bottlenecks.
- Formalised budget-constrained rewiring conditions, proving that adding shortcut hyperedges strictly reduces the Jacobian obstruction upper bound mitigating over-squashing.

ACHIEVEMENTS

GATE 2025: Secured **99.66 percentile** in the Computer Science Engineering discipline with a GATE score of 761.
IISc-Mallika Women MTech Fellowships: Awarded the prestigious fellowship for academic excellence and research potential.

EXTRACURRICULAR

Serving student representative in Gymkhana Students Affair committee - IISc – *2026, Bengaluru, Karnataka*

EECS Sports Meet Basketball Tournament – *Player* *2025, Bengaluru, Karnataka*

- Won a **Silver Medal** representing the Department of CDS.

Interface: Branch Chapter of CSE – *Technical Head* *2022 – 2025, Pantnagar, Uttarakhand*

- Organized events like PC gaming, coding competitions, and chess for the Computer Engineering Department.

National Service Scheme (NSS) – *Member* *2022 – 2024, Pantnagar, Uttarakhand*

- Contributed to community service initiatives and social awareness programs.